Problem Set 1 Solutions

QTM 220

Mean and SD Review

- Each of the following lists has a mean of 50. For which is the spread of the numbers around the mean biggest? Smallest?
  - (i) 0, 20, 40, 50, 60, 80, 100
  - (ii) 0, 48, 49, 50, 51, 52, 100 – smallest
  - (iii) 0, 1, 2, 50, 98, 99, 100 – biggest
- For each pair of lists below, calculate the mean and SD of each list, then compare within pairs.
  - (i.a) 1, 3, 4, 5, 7
  - mean = 4, SD= 2
  - (i.b) 6, 8, 9, 10, 12
  - mean= 9, SD= 2
  - How is list (i.a) related to list (i.b)? How does this relationship carry over to the mean? the SD?
  - List b is list a+5 on each entry. The mean increases by 5, but the standard deviation does not change. Adding a constant does not affect the standard deviation, but does change the mean.
  - (ii.a) 1, 3, 4, 5, 7
  - mean = 4, SD= 2
  - (ii.b) 3, 9, 12, 15, 21
  - mean= 12, SD= 6

  - How is list (ii.a) related to list (ii.b)? How does this relationship carry over to the mean? the SD?
  - List b is list a*3. The mean is multiplied by 3 and the standard deviation is also multiplied by 3. Multiplying by a (positive) constant multiplies mean and SD by that number
  - (iii.a) 5, -4, 3, -1, 7
  - mean = 2, SD= 4
  - (iii.b) -5, 4, -3, 1, -7
  - mean= -2, SD= 4
  - How is list (iii.a) related to list (iii.b)? How does this relationship carry over to the mean? the SD?
  - List b is list a with the signs flipped. The sign of the mean flips (and so do all individual deviations from the average), but does not affect the standard deviation

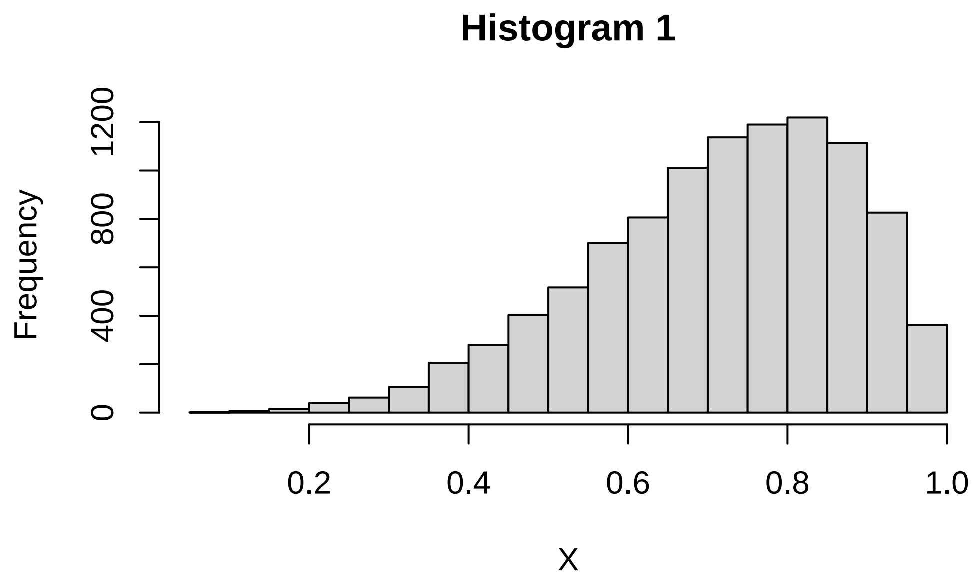
- Can the standard deviation ever be negative? Explain.

  No

- For set of positive numbers, can the standard deviation ever be larger than the average? Explain.

  Yes. Example: for a set 1,1,16, the mean is 6 and the standard deviation is about 7

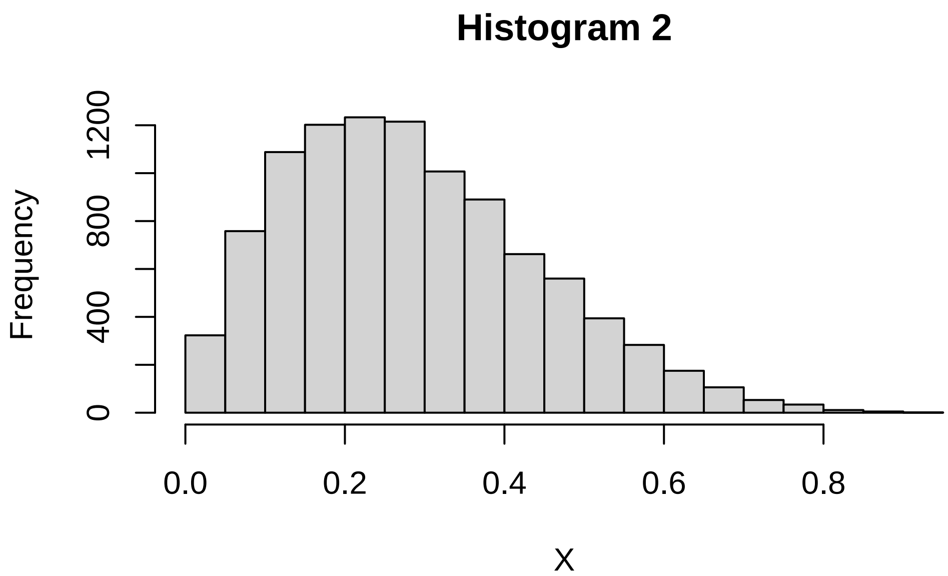
- Consider the following three histograms.
  - Match each histogram with the following approximate means: 0.3, 0.5, 0.7.
  - Match each histogram with the following descriptions: (a) the median about equal to the mean; (b) the median is larger than the mean; (c) the median is less than the mean.
  - Is the standard deviation of Histogram 2 around 0.015, 0.15, or 1.5?
  - Ture or false (and explain): the standard deviation for Histogram 1 is a lot smaller than for Histogram 2.




```
[1] 0.7169802
```

```
[1] 0.73549
```

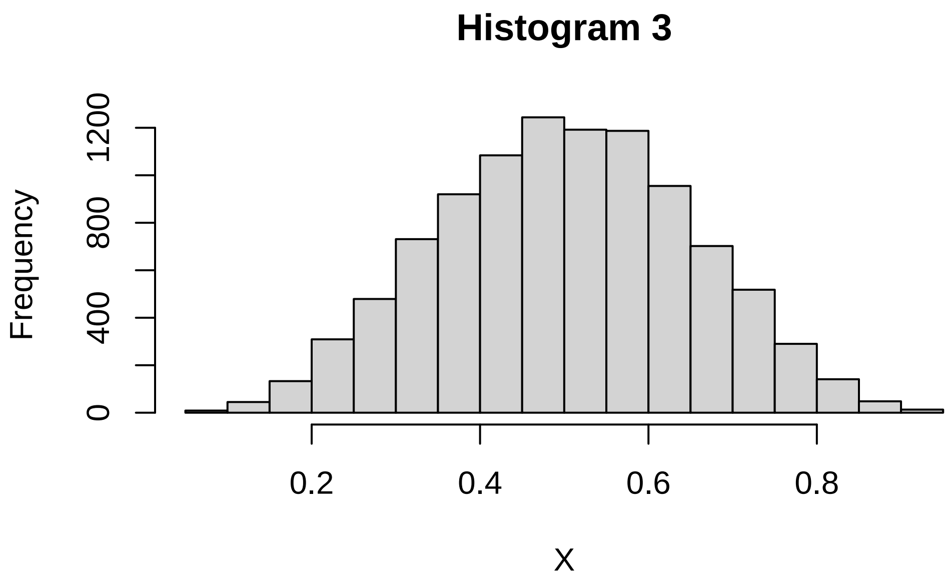
```
[1] 0.158375
```




```
[1] 0.2855042
```

```
[1] 0.2649767
```

```
[1] 0.1601153
```




```
[1] 0.5004656
```

```
[1] 0.4992844
```

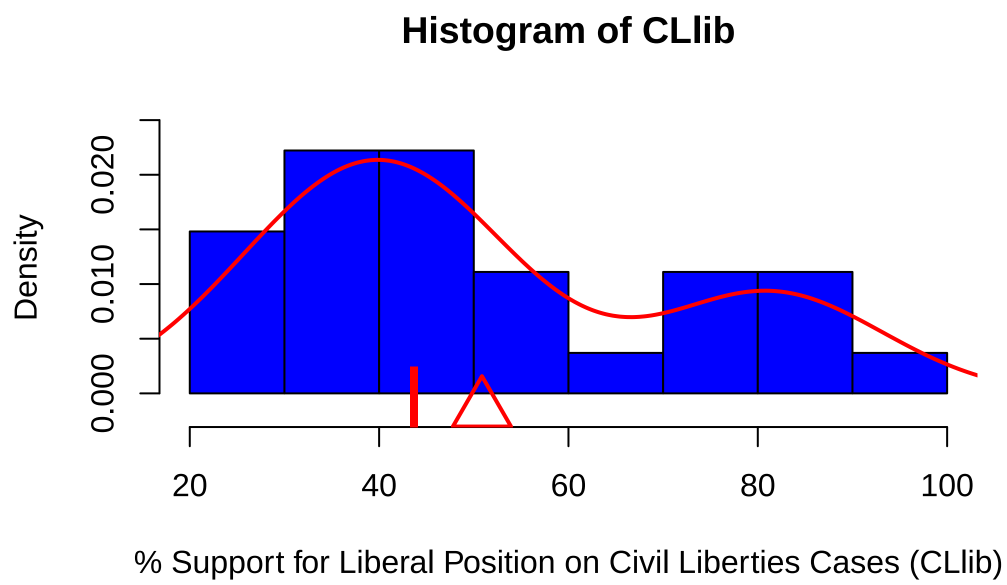
```
[1] 0.1496575
```

Supreme Court Justices

Load the provided dataset "EMdata.csv"

- This data is on 27 justices from the Warren ('53 - '69), Burger ('69 - '86), and Rehnquist ('86 - '05) courts
- The data can be interpreted as a census for the 1953 - 2005 era.
- Variables:
  - Justice: last name of Supreme Court Justice
  - CLlib: Percentage of votes in liberal direction for each justice in civil liberties cases
  - party: Political party that nominated the justice (Republican =0, Democrat=1)
  - ur: Justice is a member of an under-represented group, such as a racial or gender minority (under-represented group=1, not in under-represented group=0)

(a) Create a histogram of the percentage of liberal votes, visually identifying the mean and median.



(b) Describe what the mean and median represent in the context of the CLlib variable for these justices. Which is larger? What does that imply in this context?

SOLUTION

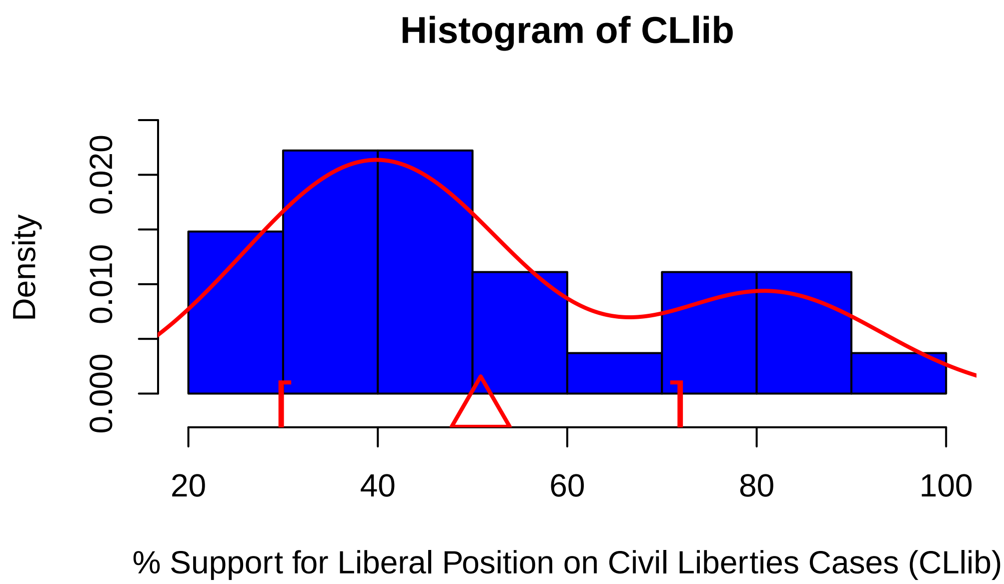
(c) Write your own function to calculate the standard deviation of CLlib (i.e. not using "sd") and report it. Add a visual identification of a standard deviation on each side of the mean.

```
y.bar<- mean(CLlib)
s.resid<- rep(0,length(CLlib))
for(i in 1:length(CLlib)){
    s.resid[i] <- (CLlib[i] - y.bar)^2
}
sqrt(sum(s.resid)/(length(CLlib)-1))
```

```
[1] 20.89227
```

```
sd(CLlib)
```

```
[1] 20.89227
```



(d) Calculate the mean of 'ur', whether a justice is a member of an under-represented group. What is the substantive interpretation of this mean?

```r
mean(ur)
```

```
[1] 0.1111111
```

The proportion of justices who are 1s, the proportion of justices who are members of an under-represented group.

(e) Calculate the conditional means and medians of CLlib conditioning on nominating party. Create two histograms of CLlib, one for Republican-nominated justices, one for Democratic-nominated justices. What do the mean and median represent in the context of the CLib variable for the justices of each party? Which is larger, the mean among Republican justices or Democratic justices? Give a substantive interpretation of this. Which is larger, the median among Republican justices or Democratic justices? Give a substantive interpreation of this. Which difference is larger? What does that mean?

```r
mean(CLlib[party==0])
```

```
[1] 44.11333
```

```r
median(CLlib[party==0])
```

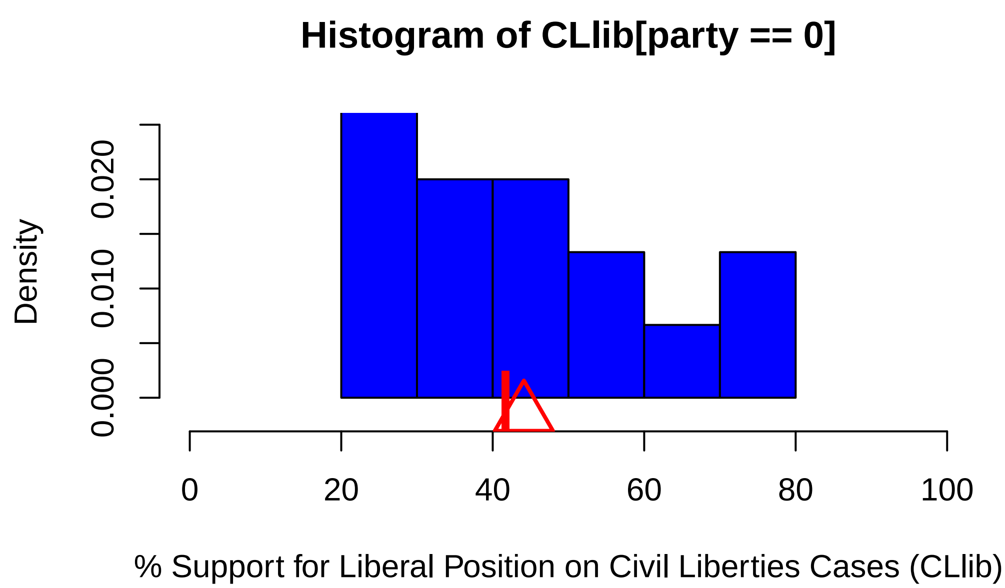
```
[1] 41.7
```

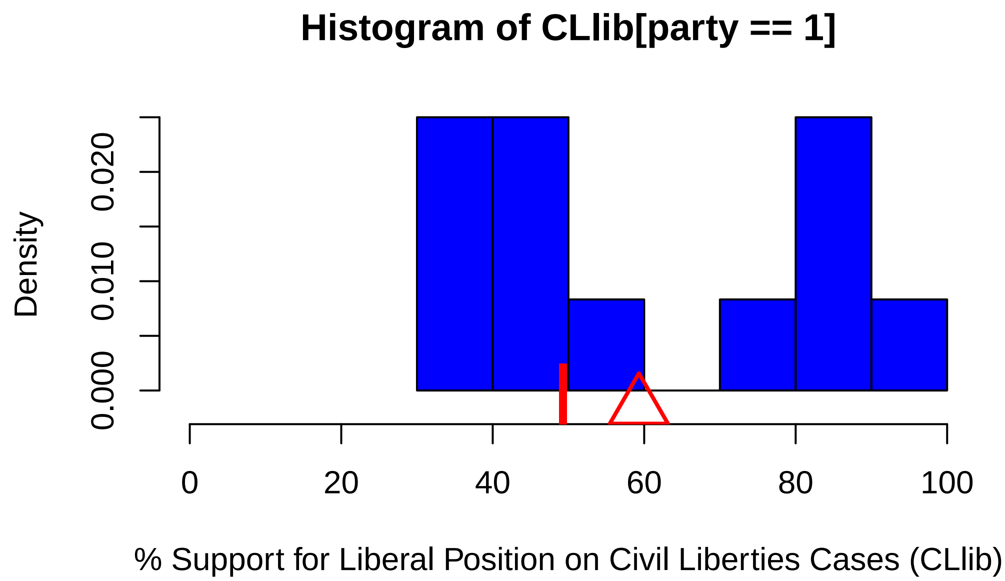
```r
mean(CLlib[party==1])
```

```
[1] 59.3
```

```r
median(CLlib[party==1])
```

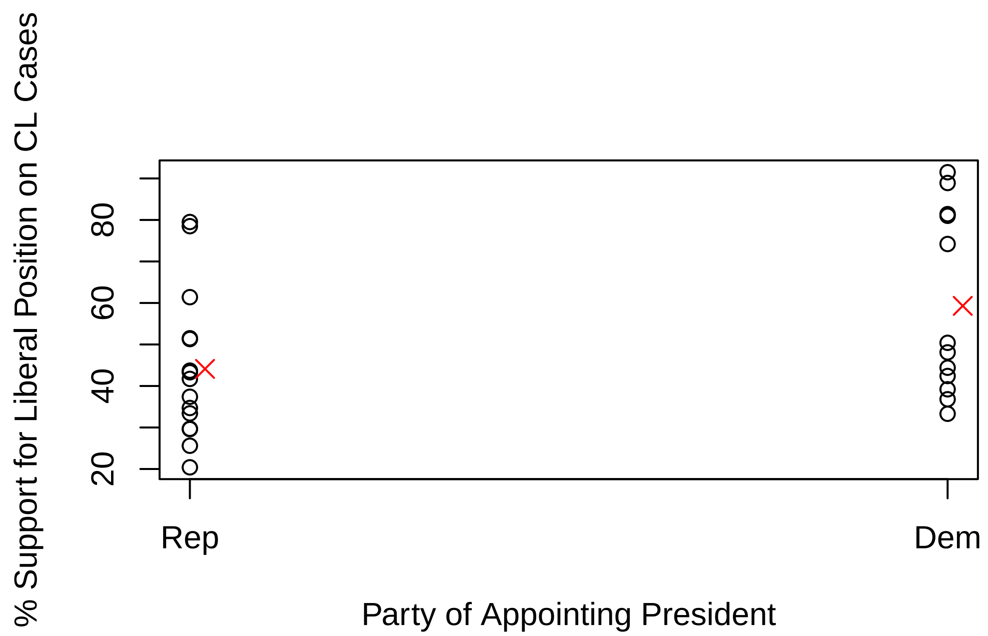
```
[1] 49.25
```





SUBSTANTIVE INT.

(f) Create a scatter plot of CLlib with the nominating party as the x-axis. Visually indicate the conditional means (conditioning on party) on the plot.

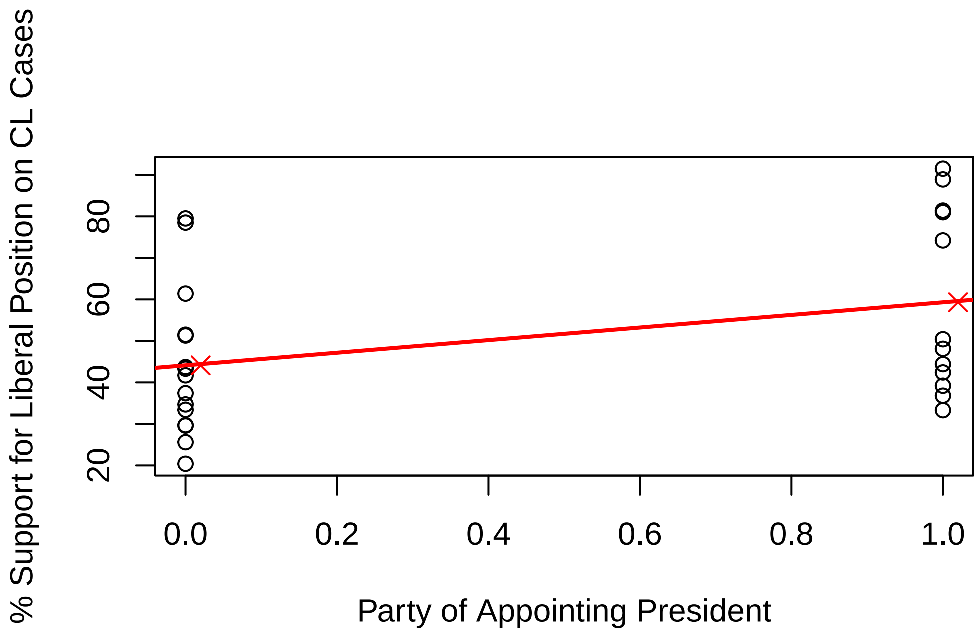


(g) Using R's lm() function, regress CLlib on party. Using abline(), add the regression line to your scatterplot. What is substantive interpretation of the intercept and slope of the line in terms of justices for each party?

```
lm(CLlib~party)
```

```
Call:
lm(formula = CLlib ~ party)

Coefficients:
(Intercept)        party
      44.11        15.19
```



SUBSTANTIVE

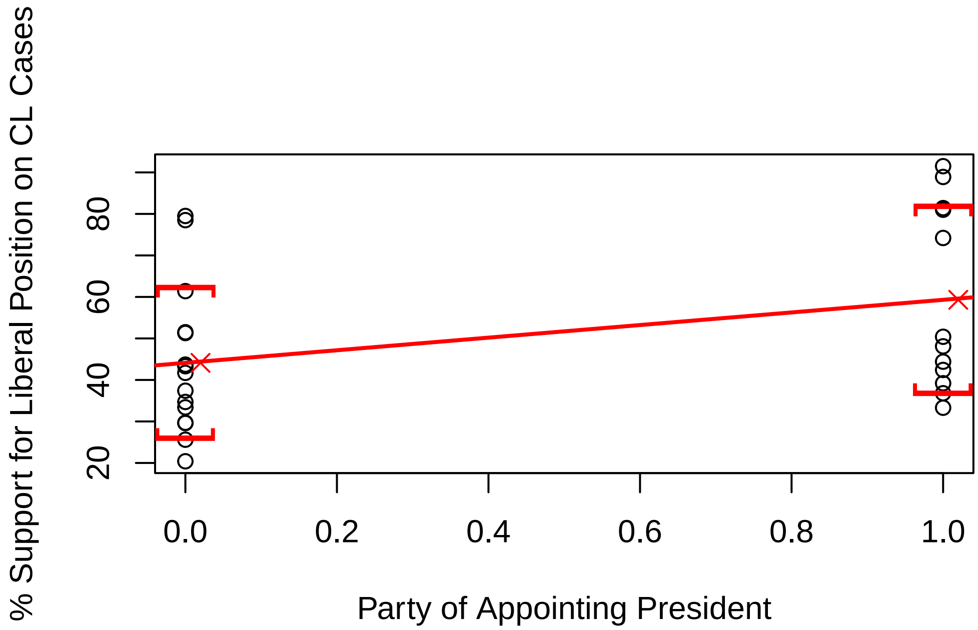
(h) Calculate the conditional standard deviations of CLlib, conditioning on party. Add a visual represenatation of the conditional standard deviations to your scatterplot.

```
sd(CLlib[party==0])
```

```
[1] 17.76485
```

```r
sd(CLlib[party==1])
```

```
[1] 22.14177
```

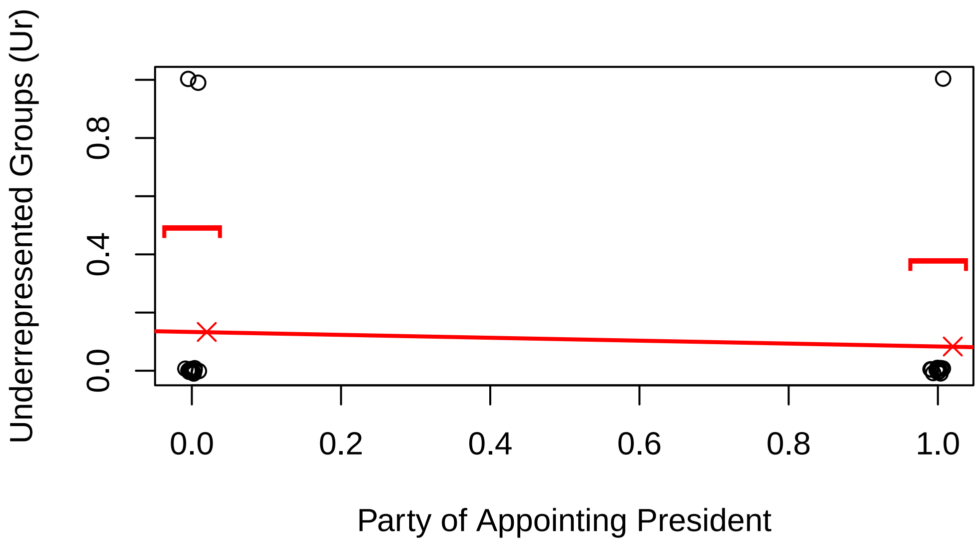


(i) Create a scatterplot of justices with under-represented group membership as the y-axis and party as the x-axis. Add the conditional means and standard deviations to the plot (conditioning on party). Using lm(), regress under-represented group membership on party and add the regression line to your plot using abline(). What is the substantive interpretation of the intercept and slope of the line in terms of justices for each party?

```r
lm(ur~party)
```

```
Call:
lm(formula = ur ~ party)

Coefficients:
(Intercept)        party
     0.1333      -0.0500
```



SUBSTANCE

Matrix Algebra and Least Squares

(a) Let Y be the nx1 vector of our outcome variable and X be the nx2 matrix that concatenates our independent variable of interest with a row of 1s. $\hat{\beta}$ is a 2x1 vector.

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$$

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix}$$

Define r as the nx1 vector of residuals as given by $Y - X\hat{\beta}$

Using matrix algebra, show that $(Y - X\hat{\beta})'(Y - X\hat{\beta})$ yields the sum of squared residuals.

$$\hat{r} = \underbrace{(Y_{nx1} - \underbrace{X_{nx2}\hat{\beta}_{2x1}}_{\text{nx1}})}_{\text{nx1}}$$

$$\hat{r}' = \underbrace{(Y_{nx1} - \underbrace{X_{nx2}\hat{\beta}_{2x1}}_{\text{nx1}})'}_{\text{1xn}}$$

$$\hat{r}' * \hat{r} = \begin{bmatrix} \hat{r}_1 & \hat{r}_2 & \hat{r}_3 & \cdots & \hat{r}_n \end{bmatrix} * \begin{bmatrix} \hat{r}_1 \\ \hat{r}_2 \\ \hat{r}_3 \\ \vdots \\ \hat{r}_n \end{bmatrix}$$

$$= \hat{r}_1 * \hat{r}_1 + \hat{r}_2 * \hat{r}_2 + \hat{r}_3 * \hat{r}_3 + \cdots + \hat{r}_n * \hat{r}_n = \sum_{i=1}^{n} \hat{r}_i^2$$

This is the sum of squared residuals. Therefore $(Y - X\hat{\beta})'(Y - X\hat{\beta})$ yields the sum of squared residuals.

(b) Using matrix algebra and $\frac{\partial}{\partial \hat{\beta}}$, show that $\hat{\beta} = (X'X)^{-1}X'Y$ minimizes the sum of squared residuals (assuming that $X'X$ is invertible.)

We want to choose $\hat{\beta}$ to minimize $(Y - X\hat{\beta})'(Y - X\hat{\beta})$, the sum of squared residuals.

$(Y - X\hat{\beta})'(Y - X\hat{\beta}) = Y'Y - \hat{\beta}'X'Y' - Y'X\hat{\beta} + \hat{\beta}'X'X\hat{\beta}$

$Y'Y - 2Y'X\hat{\beta} + \hat{\beta}X'X\hat{\beta}$

First Order Condition $\frac{\partial}{\partial \hat{\beta}} = -2X'Y + 2X'X\hat{\beta} = 0$

$-X'Y + X'X\hat{\beta} = 0$

$X'X\hat{\beta} = X'Y$

assume $(X'X)^{-1}$ exists ($X'X$ is invertible)

$\hat{\beta} = (X'X)^{-1}X'Y$

(c) Let $Y = \begin{bmatrix}1\\1\\0\\1\end{bmatrix}$ and $X = \begin{bmatrix}1 & 1\\1 & 0\\1 & 0\\1 & 1\end{bmatrix}$

Using matrix algebra, calculate $(X'X)^{-1}X'Y$.

$$X'X = \begin{bmatrix}1 & 1 & 1 & 1\\1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\1 & 0\\1 & 0\\1 & 1\end{bmatrix} = \begin{bmatrix}4 & 2\\2 & 2\end{bmatrix}$$

$$(X'X)^{-1} = \frac{1}{8-4}\begin{bmatrix}2 & -2\\-2 & 4\end{bmatrix}$$

$$(X'X)^{-1} = \begin{bmatrix}\frac{1}{2} & -\frac{1}{2}\\-\frac{1}{2} & 1\end{bmatrix}$$

$$X'Y = \begin{bmatrix}1 & 1 & 1 & 1\\1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1\\1\\0\\1\end{bmatrix} = \begin{bmatrix}3\\2\end{bmatrix}$$

$$(X'X)^{-1}X'Y = \begin{bmatrix}\frac{1}{2} & -\frac{1}{2}\\-\frac{1}{2} & 1\end{bmatrix}\begin{bmatrix}3\\2\end{bmatrix} = \begin{bmatrix}\frac{3}{2}-1\\-\frac{3}{2}+2\end{bmatrix} = \begin{bmatrix}\frac{1}{2}\\\frac{1}{2}\end{bmatrix}$$

(d) In R, create a dataframe $y = c(1, 1, 0, 1)$ and $x = c(1, 0, 0, 1)$. Calculate the conditional means of y when $x = 1$ and when $x = 0$

```
y<- c(1,1,0,1)
x<- c(1,0,0,1)
df<- data.frame(y,x)
mean(df$y[x==0])
```

```
[1] 0.5
```

```
mean(df$y[x==1])
```

```
[1] 1
```

(e) Using R's lm() function, regress y on x.

```r
lm(y~x)
```

```
Call:
lm(formula = y ~ x)

Coefficients:
(Intercept)            x
        0.5          0.5
```